

F_U_Bi vs F_U_Bi_i Distinction — Direction, Magnitude, Dimensionality

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PAPER_990: F_U_Bi vs F_U_Bi_i Distinction

Abstract

We formalize the distinction between two complementary UQFF buoyancy quantities that are frequently confused:

Property	$F_{U,Bi}$	F_{U,Bi_i}
Direction	Inside → Outside	Outside → Inside (net)
Sign	Always positive	Typically negative
Units	Energy (J)	Acceleration (m/s ²)
Magnitude	$\sim 10^{40}$	$\sim 10^{-2}$
Physical meaning	Vacuum mass transport	Net gravitational acceleration

1. F_U_Bi (Inside-to-Outside)

$$F_{U,Bi} = \rho_{SCm} \cdot V \cdot S_{26}^2 \cdot \frac{|U_b|}{|U_g| + |U_b|}$$

This is the total vacuum energy flowing outward through the 26-layer buoyancy structure. It is always positive and cosmologically large because it includes $V_{\text{region}} \sim 10^{48} \text{ m}^3$.

2. F_U_Bi_i (Outside-to-Inside)

$$F_{U,Bi_i} = U_g + U_m + U_A - U_b + F_n \cdot S_{26} \cdot \Phi \cdot E_{\text{net}}$$

This is the per-particle net acceleration through the 6-layer canonical structure. Negative values indicate net inward pull (gravity dominates buoyancy at the acceleration level).

3. Complementarity

The ratio $|F_{U,Bi_i}|/F_{U,Bi}$ gives the fractional acceleration per unit vacuum energy — the “efficiency” of gravitational focusing through the buoyancy medium.

4. Implementation

File: fubi_inside_outside.py, class FUBiDistinctionCalc. CP4 class #574.

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Jet power P_{BZ}	UQFF phonon-modulated $M_{\text{jet}}(\Gamma)$	Observed $P_{\text{jet}} \sim 10^{43}\text{--}10^{46} \text{ erg/s}$	Ghisellini et al. (2014)	Within range
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: BH-accretion (active galactic nucleus jet)

§A.2 Lagrangian Density

$$\mathcal{L}_{BH_accretion} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial (\partial_m u \phi)} = 0 \implies F_{U,Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ active galactic nucleus jet $\rightarrow F_{U,Bi_i}$ unified force $\rightarrow$ observational prediction

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\text{VDS} = \rho_{\text{SCm}} \cdot S_{26} \cdot \Phi_{1.25\text{THz}} / \Phi_0$$

VDS sub-ratio: 0.167

§B.2 Dipole Vortex Primes (DVP)

DVP prime: 73 (resonant)

§B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: $10^6 M_{\text{BH}} \text{ yr}$

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed